

CloudNova API Authentication Guide

This guide covers all authentication methods supported by the CloudNova API, including API Key authentication, OAuth2 flows, JWT token management, and troubleshooting common authentication errors.

1. API Key Authentication

API keys are the simplest way to authenticate with the CloudNova API. Each key is a unique string prefixed with 'cn_api_key_' followed by 40 alphanumeric characters.

How to generate an API Key:

1. Log in to the CloudNova Console
2. Navigate to Profile > API Keys > Generate New Key
3. Select the scope: read-only, read-write, or admin
4. Set an expiration period (90 days recommended)
5. Copy the key immediately (it will not be shown again)

Include the API key in the Authorization header:

Authorization: Bearer cn_api_key_XXXXXXXXXXXXXXXXXXXXXX

API Key Best Practices:

- Never hardcode API keys in source code
- Use environment variables or secret management tools
- Rotate keys every 90 days
- Use scoped keys with minimum required permissions
- Revoke compromised keys immediately at Profile > API Keys > Revoke

2. OAuth2 Authentication

CloudNova supports OAuth2 for applications that need to act on behalf of users.

Supported Grant Types:

- Client Credentials: Server-to-server authentication
- Authorization Code: User-delegated access with consent
- Authorization Code with PKCE: For mobile and SPA applications

Client Credentials Flow:

1. Register your application at Settings > OAuth Applications
2. Obtain client_id and client_secret
3. Request a token:

POST <https://auth.cloudnova.io/oauth/token>

Content-Type: application/x-www-form-urlencoded

grant_type=client_credentials

&client_id=YOUR_CLIENT_ID

&client_secret=YOUR_CLIENT_SECRET

&scope=compute:read compute:write

4. Use the returned access_token in API requests
5. Token expires in 3600 seconds (1 hour)

3. JWT Token Structure

CloudNova issues JSON Web Tokens (JWT) for authenticated sessions. Each token contains three parts: Header, Payload, and Signature.

Token Payload Fields:

- sub: User or service account ID
- iss: <https://auth.cloudnova.io>
- aud: <https://api.cloudnova.io>
- exp: Expiration timestamp (Unix epoch)
- iat: Issued-at timestamp
- scope: Space-separated list of granted scopes
- org_id: Organization identifier

Token Validation:

- Verify the signature using CloudNova's public JWKS endpoint
- JWKS URL: <https://auth.cloudnova.io/.well-known/jwks.json>
- Check that 'exp' has not passed (tokens are valid for 1 hour)
- Verify 'iss' matches <https://auth.cloudnova.io>
- Verify 'aud' matches your API endpoint
- System clock skew tolerance: 30 seconds

4. Common Authentication Errors

Error 401 - INVALID_API_KEY:

Cause: The API key is malformed, expired, or revoked.

Fix: Generate a new API key at Profile > API Keys.

Error 401 - TOKEN_EXPIRED:

Cause: The JWT access token has expired.

Fix: Request a new token using your refresh token or client credentials.

Note: Tokens expire after 1 hour. Implement automatic token refresh.

Error 401 - INVALID_SIGNATURE:

Cause: The JWT signature verification failed.

Fix: Ensure you're using the correct JWKS endpoint for verification.

Check that your system clock is synchronized (NTP).

Error 401 - CLOCK_SKEW:

Cause: Server time differs from your system by more than 30 seconds.

Fix: Synchronize your system clock using NTP.

Linux: `sudo ntpdate pool.ntp.org`

Windows: `w32tm /resync`

Error 403 - INSUFFICIENT_SCOPE:

Cause: The token does not have the required scope for this operation.

Fix: Request a new token with the appropriate scopes.

Available scopes: compute:read, compute:write, storage:read, storage:write, billing:read, admin:full

Error 429 - RATE_LIMITED:

Cause: Too many authentication requests.

Fix: Implement exponential backoff. Cache tokens and reuse until expiry.

Auth rate limit: 10 requests per second per client.

5. Security Best Practices for API Authentication

1. Use environment variables for credentials:

```
export CLOUDNOVA_API_KEY=cn_api_key_xxxx
```

Never commit credentials to version control.

2. Implement token caching:

Cache access tokens and reuse until close to expiry.

Refresh tokens 5 minutes before expiration.

3. Use the principle of least privilege:

Request only the scopes your application needs.

Use read-only keys for monitoring and reporting.

4. Monitor for unauthorized access:

Enable CloudTrail for API authentication events.

Set up alerts for authentication failures.

Review API key usage at Profile > API Keys > Usage.

5. Implement automatic key rotation:

Use CloudNova's key rotation API to programmatically rotate keys.

```
POST /api-keys/{key-id}/rotate
```

The old key remains valid for 24 hours after rotation.

6. Use IP allowlists for API keys:

Restrict API key usage to specific IP ranges.

Configure at Profile > API Keys > IP Restrictions.

For additional help, contact support@cloudnova.io or visit

<https://docs.cloudnova.io/authentication>